

Lecture 12

ChE 467/596-012 Polymer Rheology

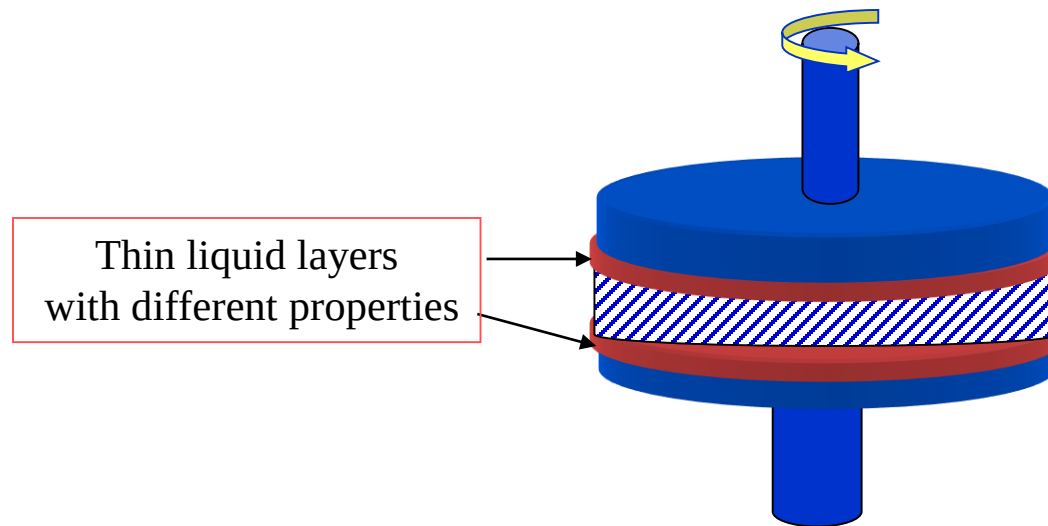
Saad A Khan

Rheometry-Wall Slip

Wall slip phenomena

Major issue with materials such as: filled systems, hydrogels, foams, emulsions

Thin film develops near walls; properties of film different from bulk



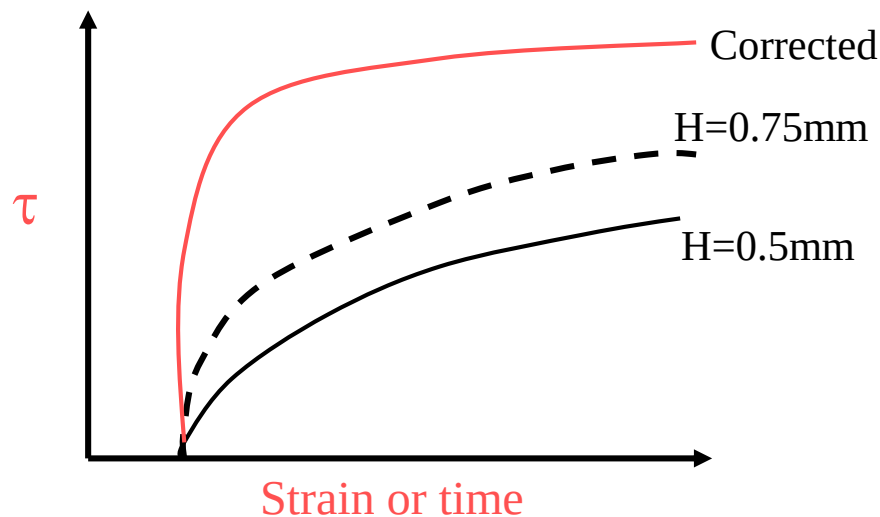
Rheometry-Wall Slip, cont'd

Test for wall slip (P&P):

Do measurements at 2 different sample thickness (gap size)

same results => no wall slip

Get different results => wall slip present



REF: Yoshimura
& Prud'homme

Rheometry-Wall Slip, cont'd

Couette system

Test for wall slip:

Repeat experiments with 2 different radii of bob (inner cylinder) keeping the R/L ratio the same.

When slip present, can calculate slip velocity:

$$v_s = \frac{\kappa}{\kappa + 1} \left[\frac{\Omega_1 - \Omega_2}{\frac{1}{R_1} - \frac{1}{R_2}} \right]$$

Ω_i =rotation rate; R_i =inner cylinder radius

$$\kappa = R_i / R_o$$

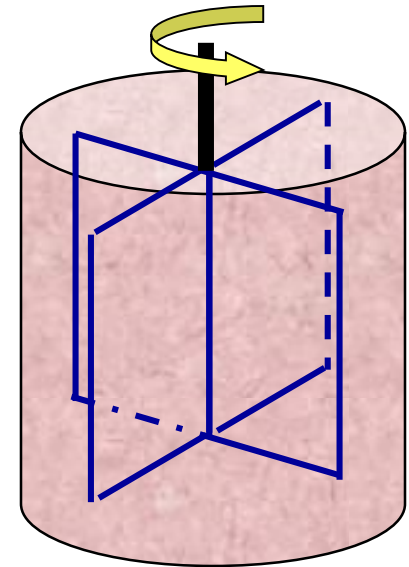
Possible ways to eliminate wall Slip

Parallel Plate system

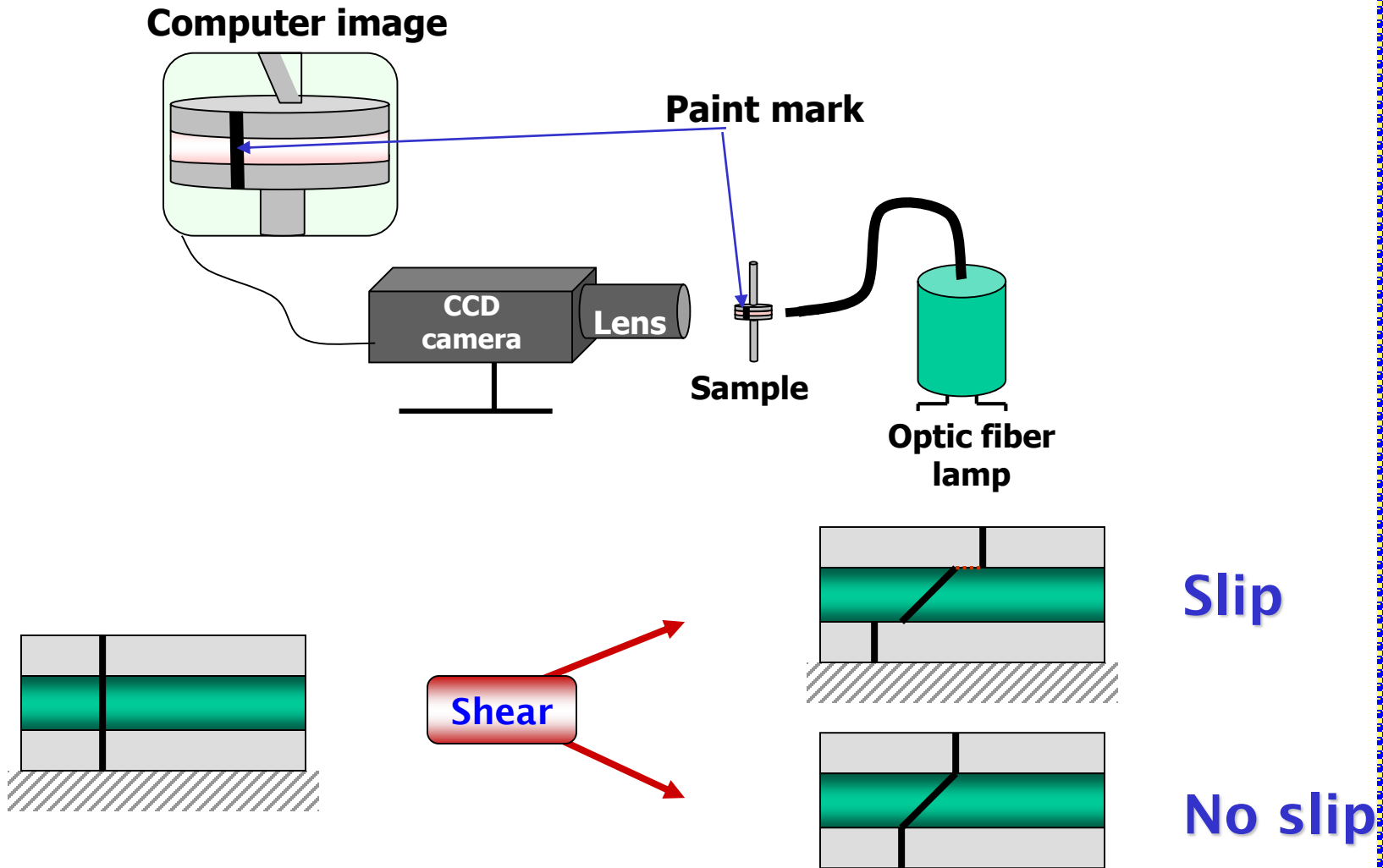
- Use serrated plates
- Glue sand paper on plates
- Do measurements at diff. gap and use correction

Couette system:

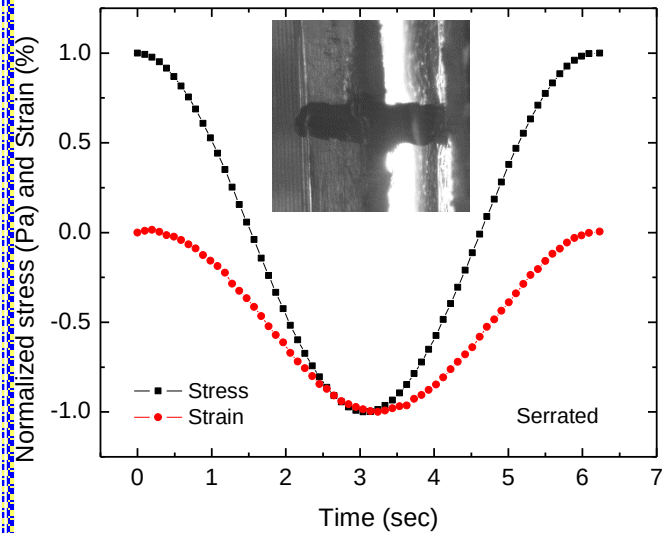
- Repeat experiments with 2 different radii of bob (inner cylinder) and use correction
- Replace inner cylinder with vane geometry.
--Useful for measuring yield stress and eliminating wall slip



Flow visualization of wall slip

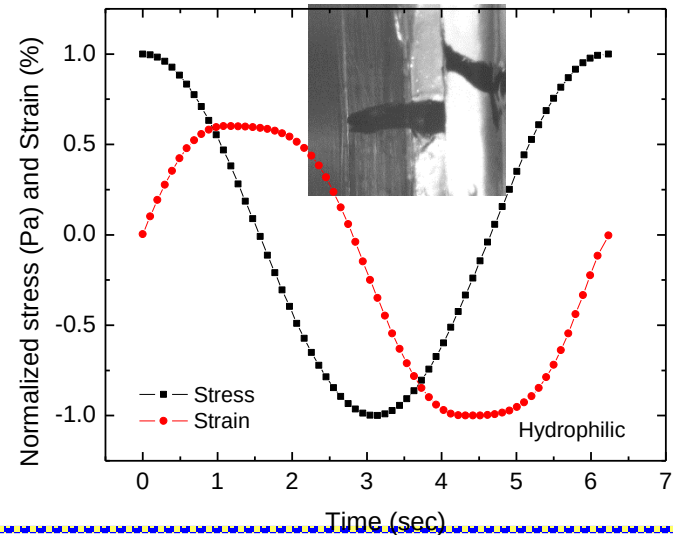


Flow visualization and rheology



No slip

Slip occurring



Modes of rotational rheometers

Strain controlled

- apply strain rate/strain => measure stress
- viscosity versus shear rate
- G' , G'' vs. frequency at const strain amplitude

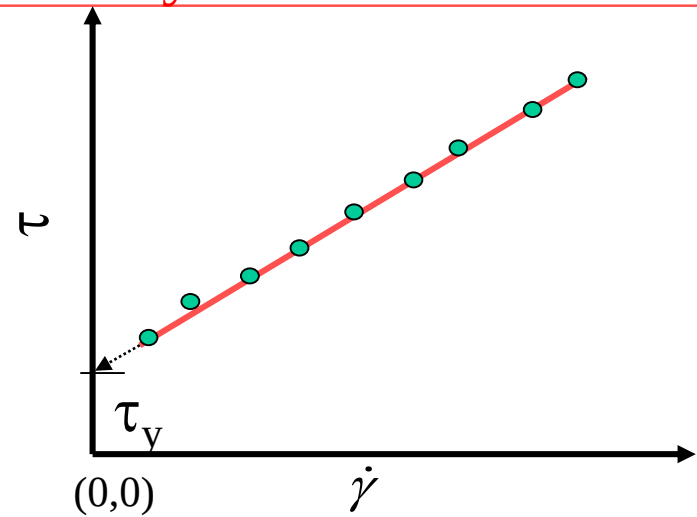
Stress controlled

- apply stress => measure strain
- viscosity versus shear stress
- G' , G'' vs. frequency at const stress amplitude
- Creep

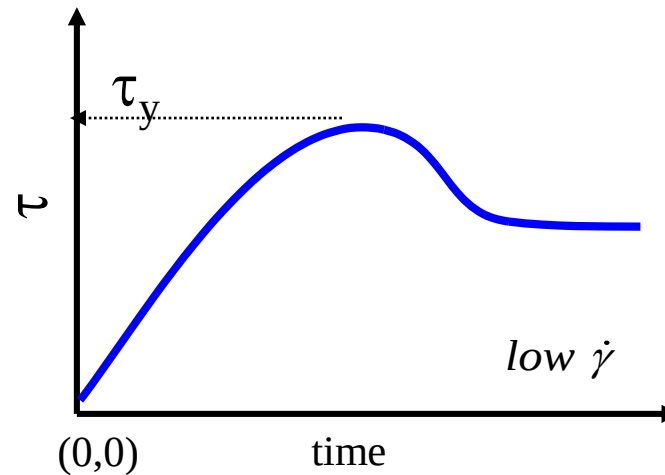
Yield Stress (τ_y)

Ways to measure τ_y :

1. Steady shear experiment



2. Vane geometry:

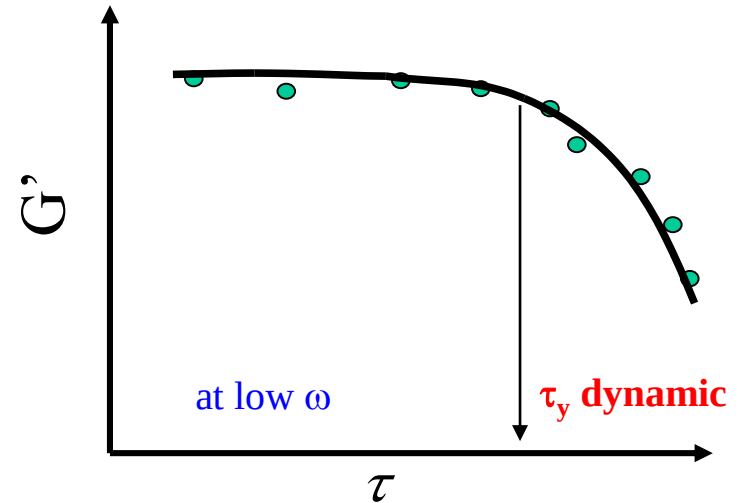


Yield Stress-cont'd

Non-traditional approaches

3. Dynamic stress sweep

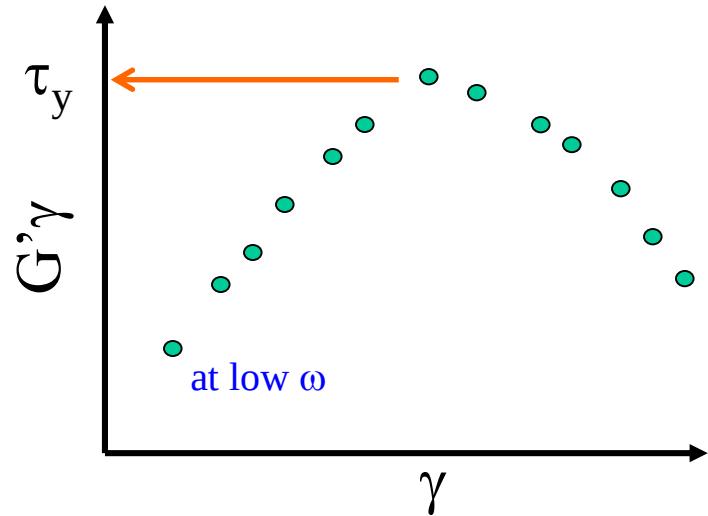
Concept of structure breakdown
as in colloidal system



4. Dynamic strain sweep:

Concept of elastic stress, $G'\gamma$

Maximum in elastic stress $\sim$
yield stress



Yield Stress-cont'd

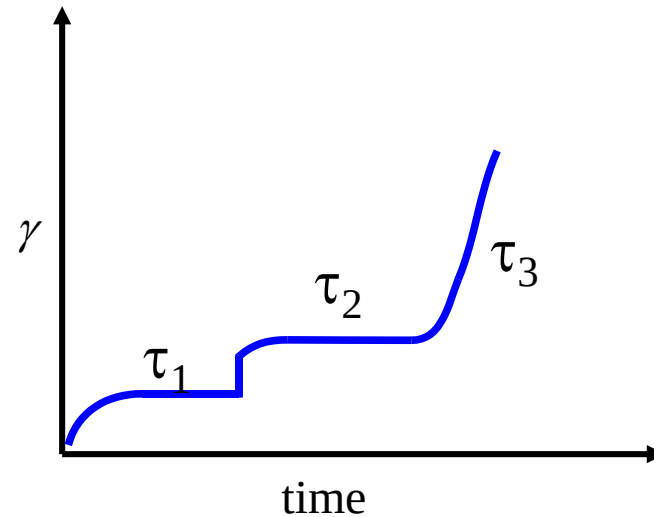
5. Creep experiment:

**Apply stress τ_1 and
measure deformation
(strain)**

**Increase applied stress in
steps until sample starts to
deform**

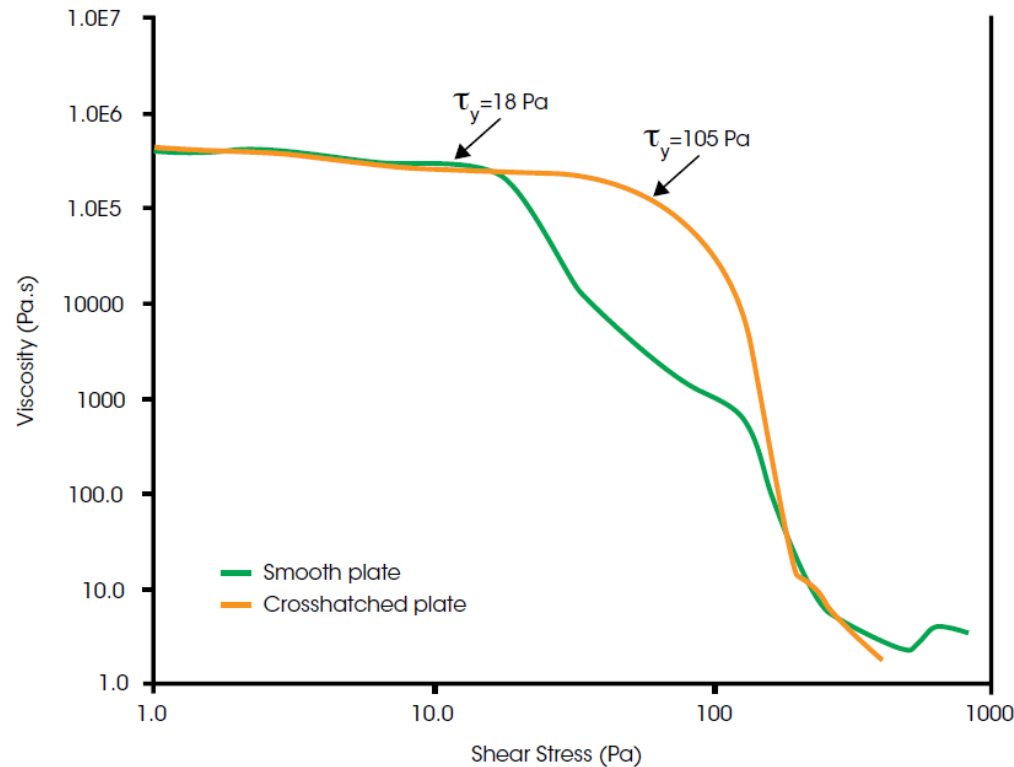
τ_y is between τ_2 and τ_3

**Do small steps between τ_2
and τ_3 pinpoint τ_y
accurately**

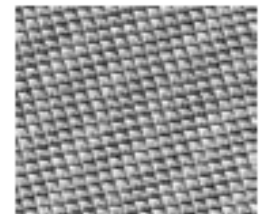


Wall Slip

Yield Stress Measurements on Toothpaste



Sandblasted

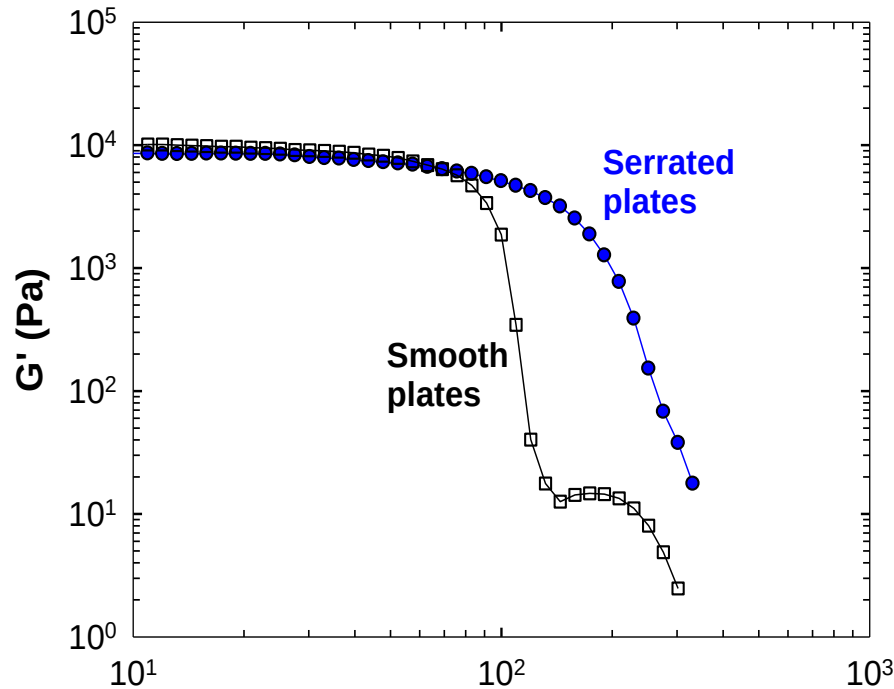


Crosshatched

- Wall slip can manifest as “apparent double yielding”
- Can be tested by running the same test at different gaps

For samples that don't slip, the results will be independent of the gap

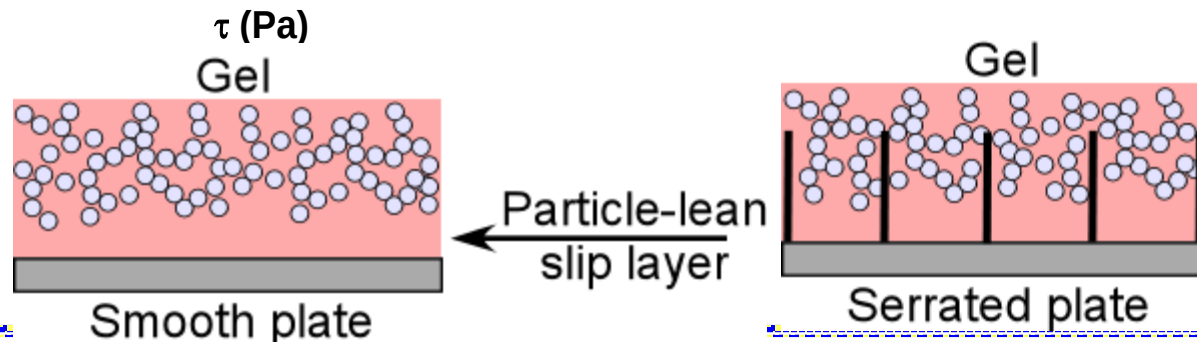
Yield stress and wall slip –actual data



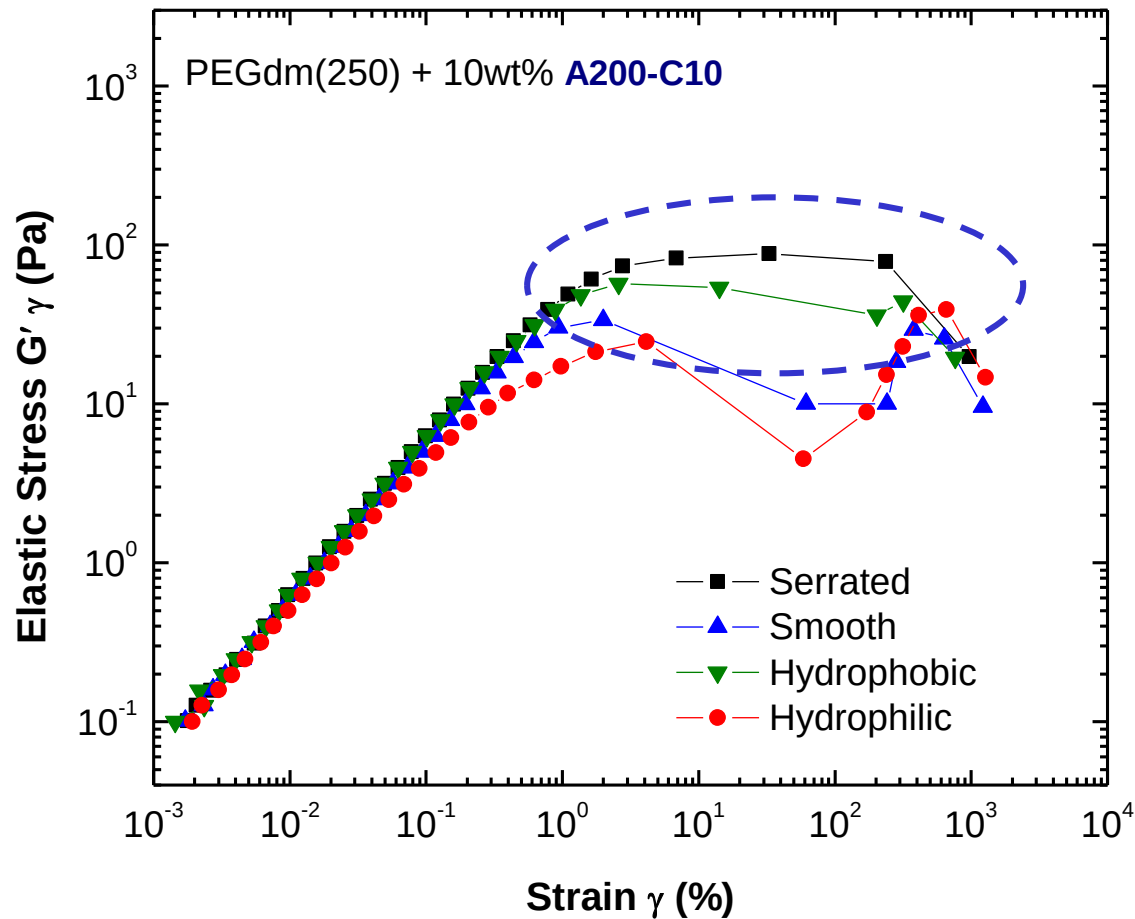
PEGdm(250) + 10wt% hydrophobic FS^(*)

$T = 25\text{ }^{\circ}\text{C}$, $w = 1\text{ rad/s}$

^(*)H. Wallset *et al. J. Rheol.* **47**(4), (2003)



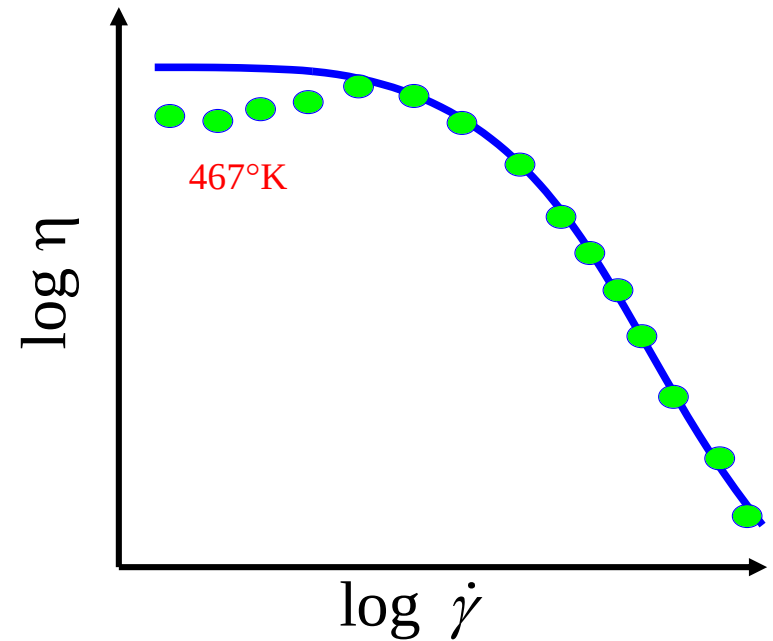
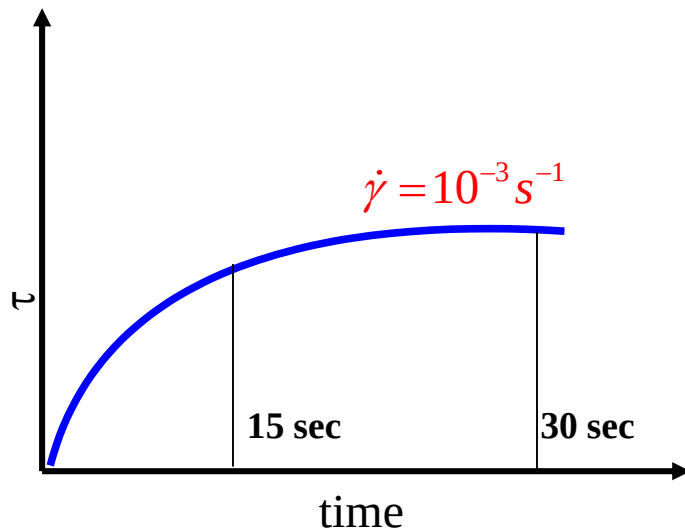
Elastic stress method



In –class exercise

What are possible causes for discrepancy at low shear rates?

Are the green data at low shear rates reliable?



- Sample not reaching steady state with green data
- Increase waiting time before measurement (e.g., from 15 to 30s)

Old exam

#1. (10) *True or False*

- (a) Newtonian fluids exhibit no normal stress difference in any flow (e.g. shear, extension).
- (b) The zero shear viscosity (η_0) of polymers can also be obtained from dynamic oscillatory shear experiments.
- (c) The steady state extensional viscosity at low extension rates is 2 times the zero shear viscosity.
- (d) Shear thinning, or for that matter shear thickening, viscosities are examples of non-linear (i.e. stress not proportional to strain or strain rate) rheological behavior.
- (e) A Bingham plastic model could be used to fit Newtonian fluid data.

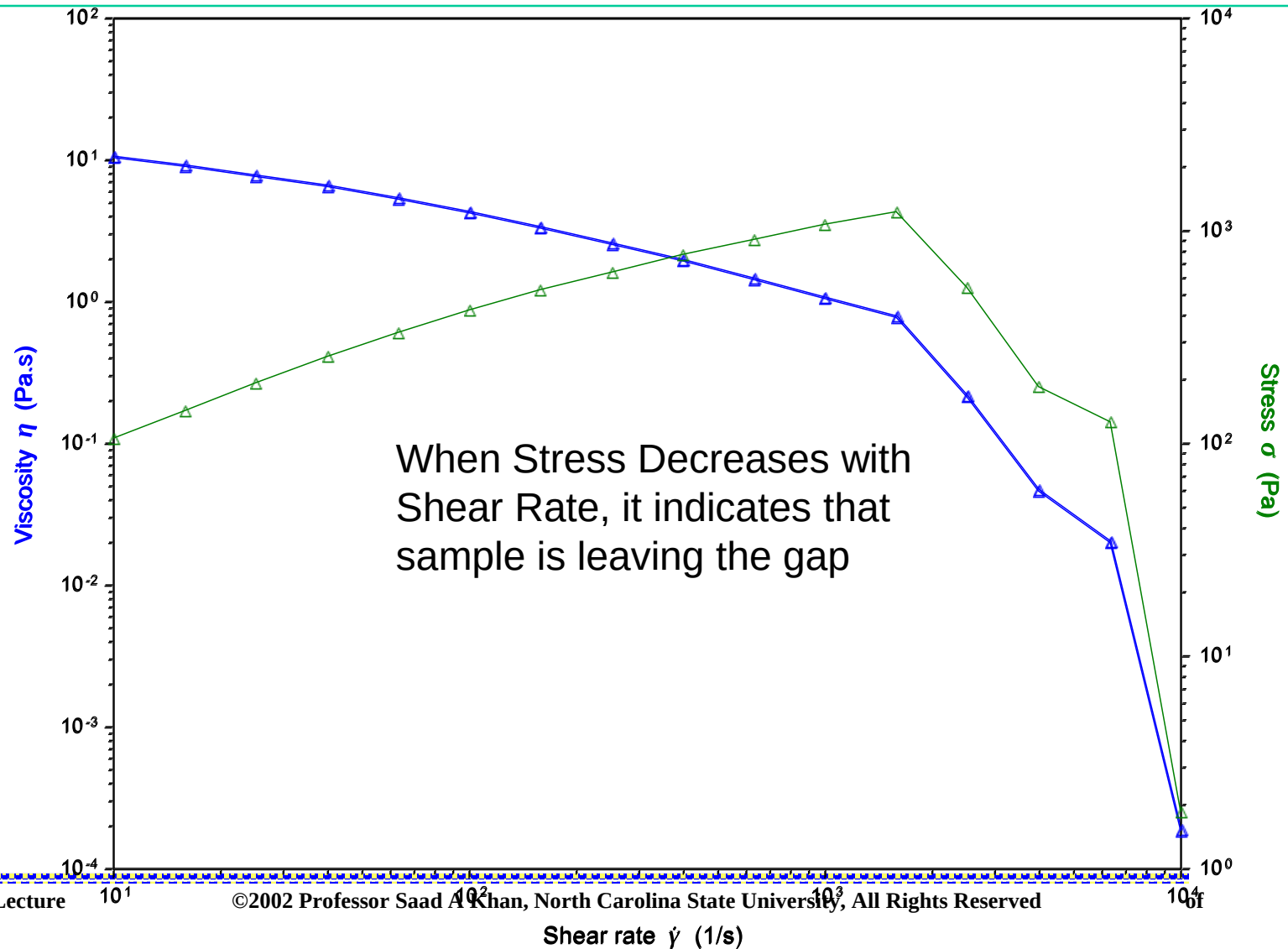


More exam questions!

True or False – you can provide justification to your answer

- (a) Normal stress generated by a polymer can be measured in a capillary rheometer by embedding pressure transducers along the wall of the capillary.
- (b) Consider a polymer melt for which G' , G'' vs ω is measured within the linear viscoelastic regime at two different strain amplitudes. The results are going to be different for the two cases.
- (c) Spinning a spindle in a beaker full of honey will exhibit dramatic rod climbing effect.
- (d) For a process in which the Deborah number > 100 , one would expect the material to behave more like a solid.
- (e) You would expect the dot product of the tensors and vectors $\underline{\underline{A}} \cdot \underline{\underline{B}} \cdot \underline{n}$ to be a scalar.
- (f) In a step-strain experiment conducted for a polymer melt, the damping function h is zero for experiments conducted within the linear (viscoelastic) regime.

Shear Thinning---Data Interpretation



Recap & Class Agenda

Last few lectures

- Examined rotational rheometers
 - advantages and pitfalls of different geometries
 - measurement techniques

Today and next class

- Old exam problems
- More analysis and pitfalls
- Pressure driven flow
 - capillary rheometer

-- analyze data

Exam Problem #1

- **Problem 1** (18) *True or False* (you can provide justification if you want)
- Rod climbing effect observed for polymeric liquids is often attributed to normal stresses.
- In shear flow, the distance between neighboring particles changes very slowly with time.
- Extensional flows are sometimes called strong flows whereas shear flow are often referred to as common flow.
- Die swelling during polymer extrusion decreases with increasing L/D ratio of the die.
- Dynamic experiments have to be done at very small strain amplitude to avoid equipment damage.
- The plateau modulus of a polymer is independent of molecular weight and temperature.



Yield Stress Measurements Revisited

Use of phase angle? Youtube video!

<https://www.youtube.com/watch?v=dz8TTrTCCjw&feature=youtu.be>

Does this relate to any of the methods we talked about?

$$G' = \frac{\text{stress amplitude}}{\text{strain amplitude}} \cos \delta$$

Inertial Effects

- What is Inertia?
 - Definition: That property of matter which manifests itself as a resistance to any change in momentum of a body
 - Instrument has inertia
 - Sample has inertia

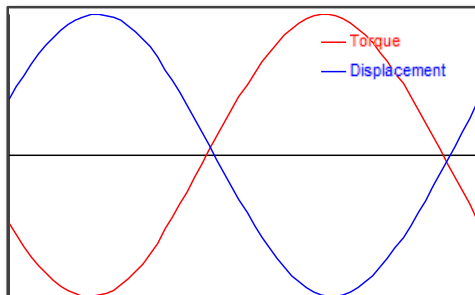
Inertial Effects in Oscillation for DHR

- Inertia consideration
 - G' , G'' limitations with frequency
 - Minimize inertia by using low mass geometries
 - Monitor inertia using Raw Phase in degree
 - When Raw Phase very large (>150 deg)
 - This indicates that the system inertia is dominating the measurement signal. Data may not be valid

Raw Phase $\times$ Inertia Correction = delta

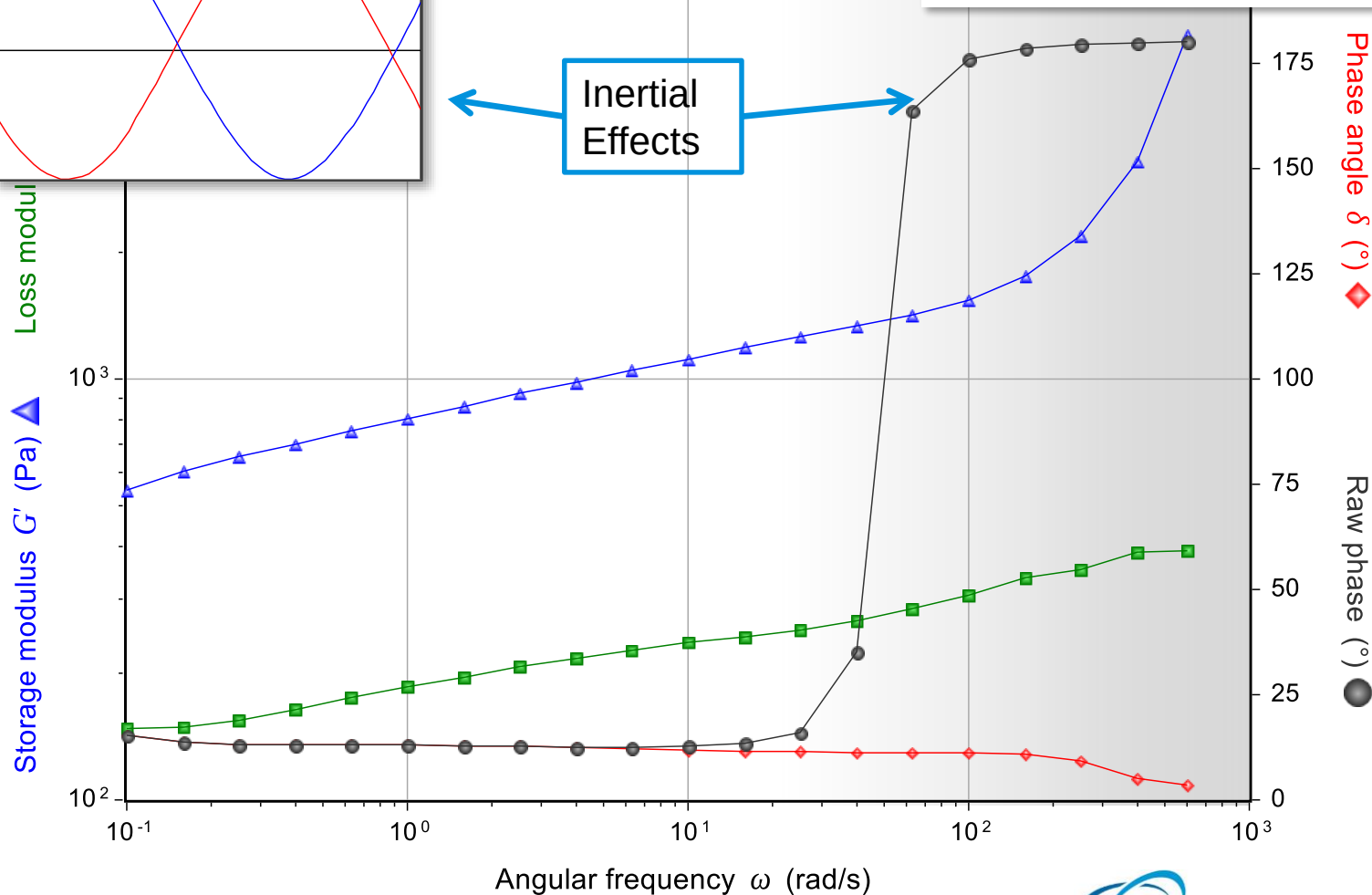
DHR Correction for Inertia

Waveforms at high frequencies



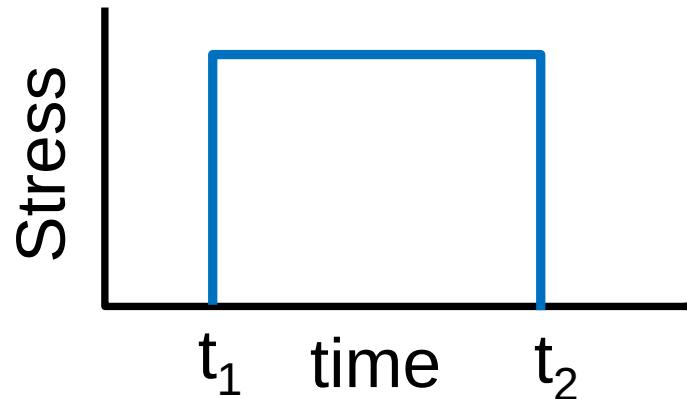
Hand Cream

Access to raw phase angle
only available with TA
Instruments Rheometers!

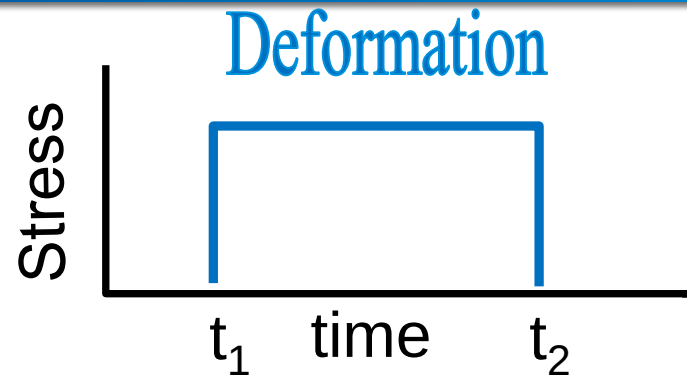


Creep Recovery Experiment

- Stress is applied to sample instantaneously, t_1 , and held constant for a specific period of time. The strain is monitored as a function of time ($\gamma(t)$)
- The stress is reduced to zero, t_2 , and the strain is monitored as a function of time ($\gamma(t)$ or $\varepsilon(t)$)



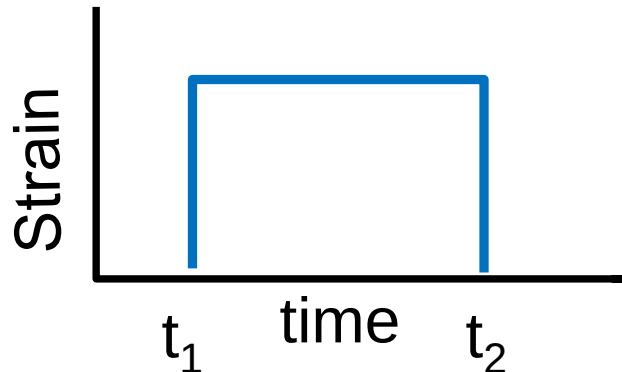
Creep Recovery Experiment



Response of Classical Extremes

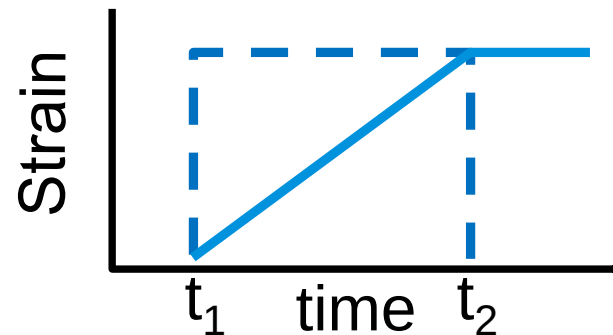
Elastic

- Strain for $t > t_1$ is constant
- Strain for $t > t_2$ is 0

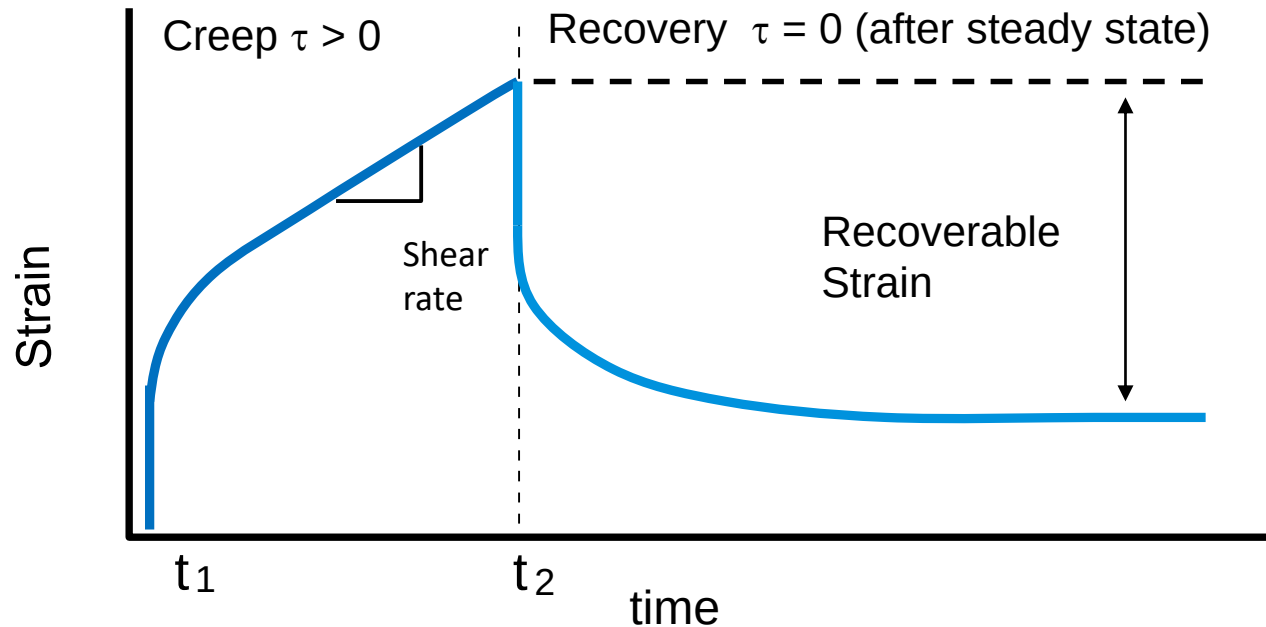


Viscous

- Strain rate for $t > t_1$ is constant
- Strain for $t > t_1$ increase with time
- Strain rate for $t > t_2$ is 0



Creep Recovery: Response of Viscoelastic Material

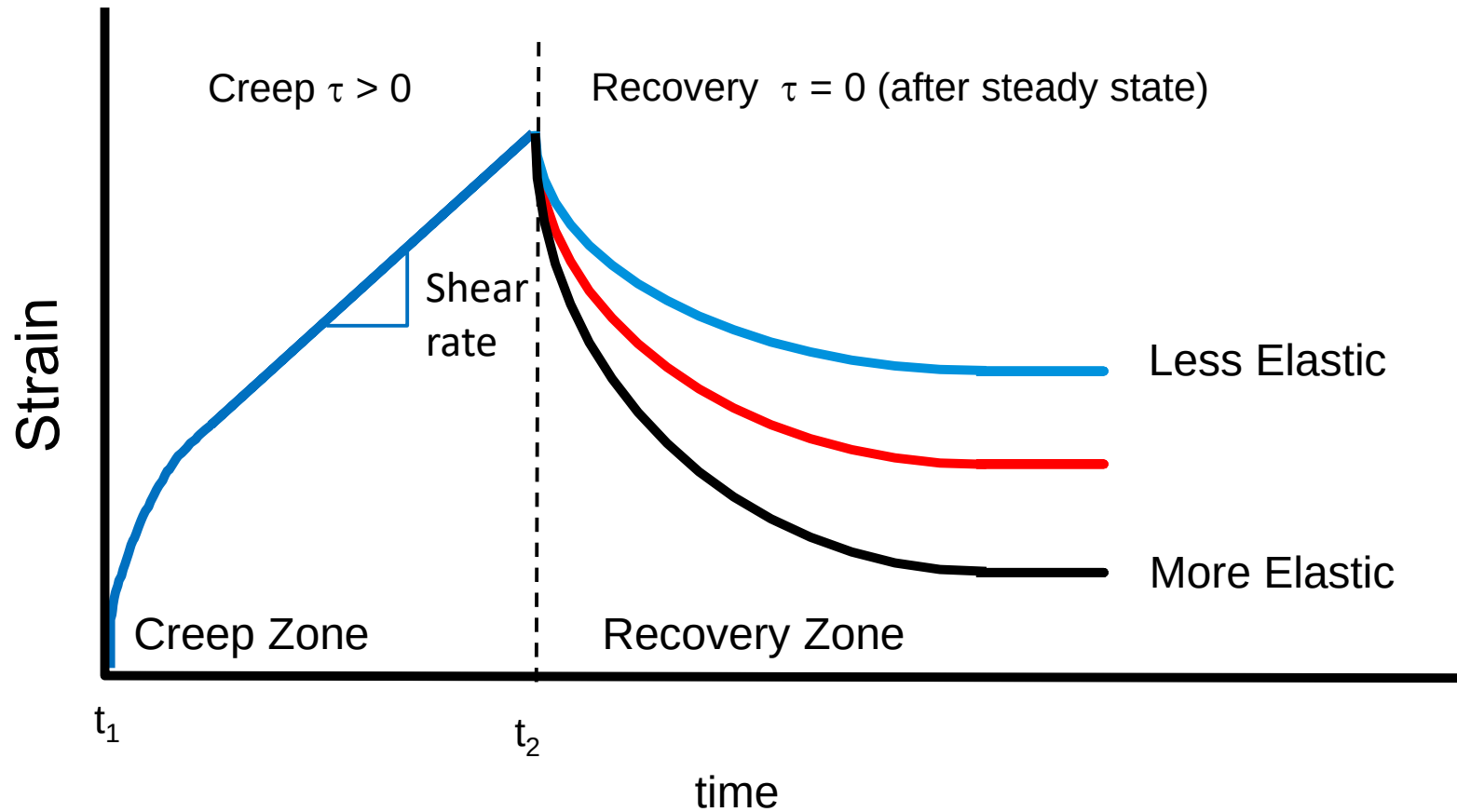


Strain rate decreases with time in the creep zone, until finally reaching a steady state.

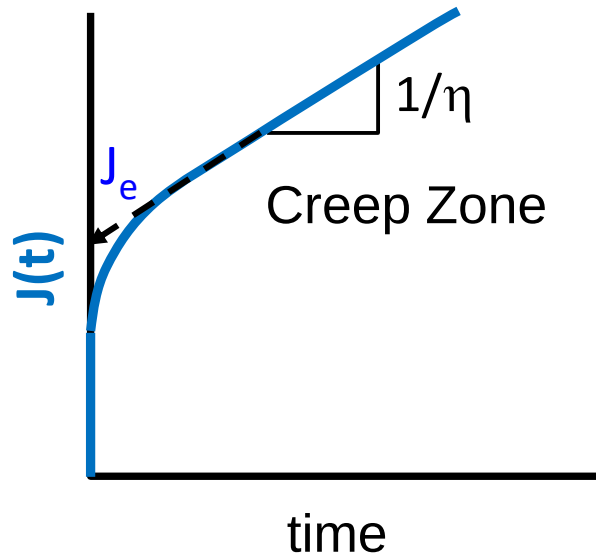
In the recovery zone, the viscoelastic fluid recoils, eventually reaching a equilibrium at some small total strain relative to the strain at unloading.

Mark, J., et. al., Physical Properties of Polymers, American Chemical Society, 1984, p. 102.

Creep Recovery Experiment



Creep Recovery : Creep and Recoverable Compliance

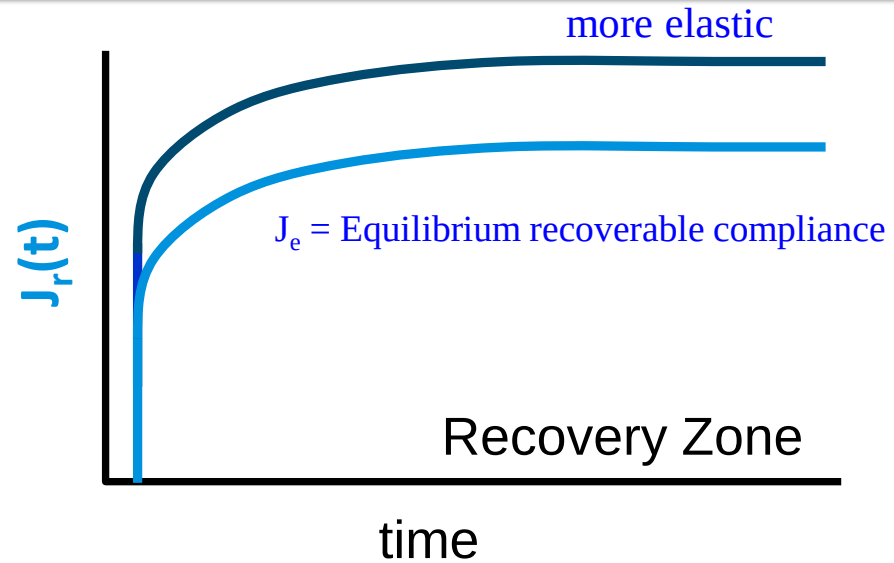


Creep Compliance

$$J(t) = \frac{\gamma(t)}{\tau}$$

The material property obtained from Creep experiments:

Compliance = 1/Modulus (in a sense)



Recoverable Compliance

$$J_r(t) = \frac{[\gamma_u - \gamma(t)]}{\tau}$$

Where

γ_u = Strain at unloading

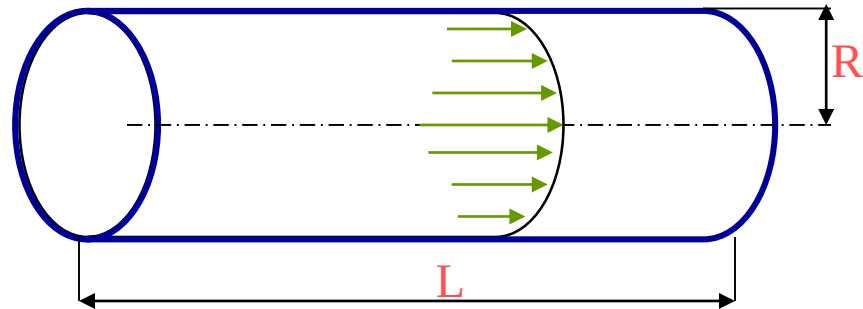
$\gamma(t)$ = time dependent recoverable strain

Mark, J., et. al., Physical Properties of Polymers, American Chemical Society, 1984, p. 102.

Pressure Driven Flow (Capillary Rheometer)

Principle => pipe flow

Non-homogenous flow



Measure shear stress at wall:

$$\tau_{wall} = \tau_{12} \Big|_{r=R} = \frac{R}{2} \frac{\Delta p}{L}$$

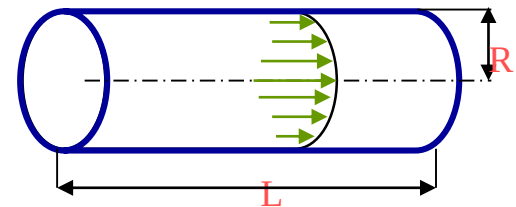
Measure flowrate:

$$\dot{\gamma}_{wall} = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{wall}} \right]$$

$$\eta = \frac{\tau_{wall}}{\dot{\gamma}_{wall}}$$

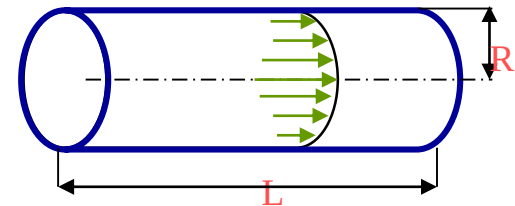
Capillary Rheometers-Disadvantages

- Non-homogenous flow (Correction factor needed)
- High shear rates only
- Difficult to Clean (sample sticks on wall, solidification)
- Can Only measure viscosity
 - No normal stresses
 - No dynamic properties (G' , G'' , etc)
- Compressibility issues
- Viscous heating
- Pressure effect on η



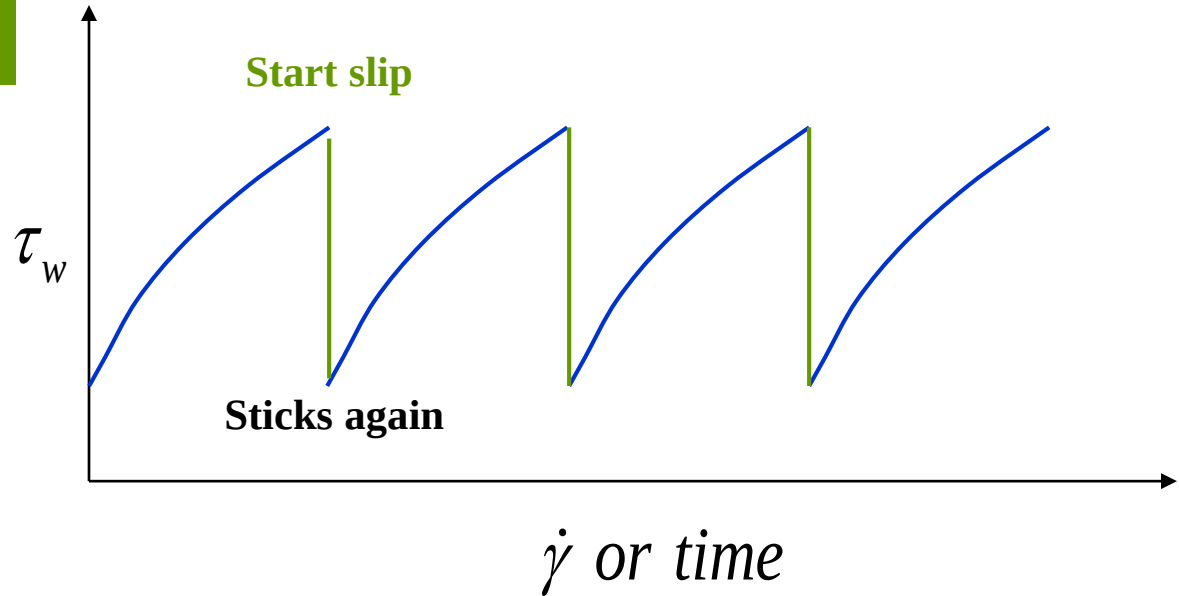
Capillary Rheometer-Advantages

- No sample evaporation (Temperature studies)
- Can get high shear rate data
 - Relevant in processing
 - Expands the shear rate regime
- Similar to process flow (can have it as online device)
- Enclosed (good for volatile materials)
- Can use high pressure



Cappillary rheometry –other issues

Stick Slip



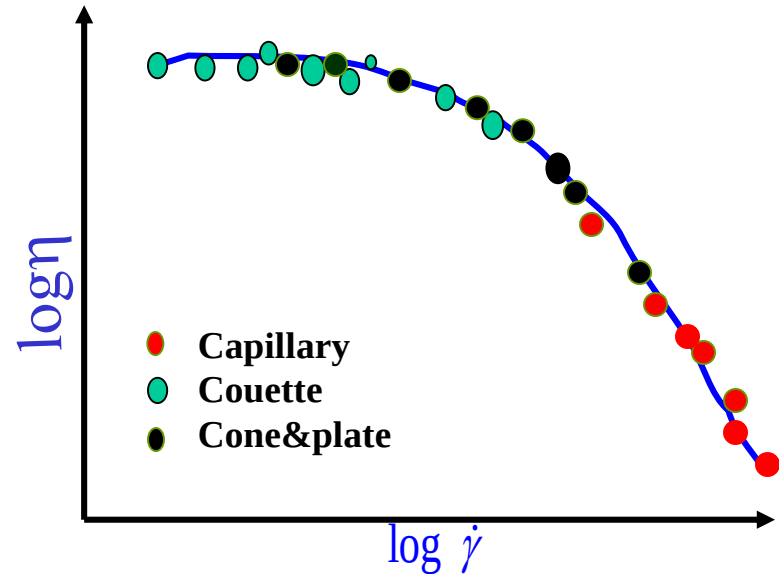
Melt Fracture:

High shear rates $\rightarrow$ unstable flow $\rightarrow$
extrudate appears distorted

Jagged polymer samples coming out

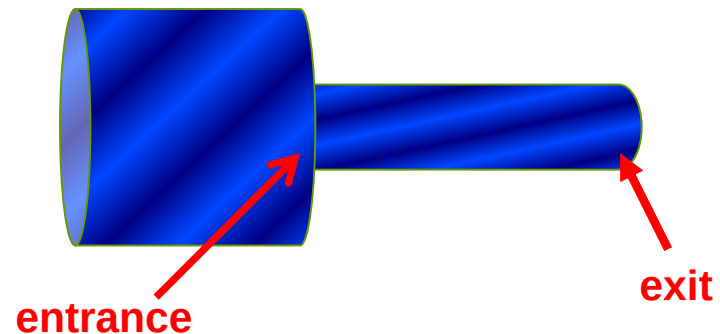
Capillary Rheometry-Other Issues

- Data can be combined with data from couette and cone & plate rheometers to get wide shear rate range



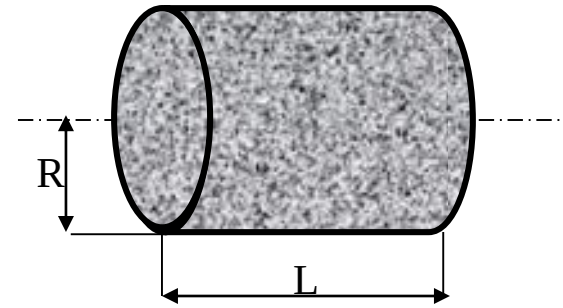
Wall slip – need to correct shear rate

Entrance/exit effects
-need to correct stress



Wall Slip Effects

- Wall slip can happen, particularly if you have filled systems
- Fillers migrate inwards → particle lean layer near wall with much lower viscosity → slip layer



To check for wall slip:

Measure viscosity at similar L/R ratio but different R

If results overlap => no wall slip

If different results => wall slip present

Wall Slip Effects, cont'd

Correcting shear rate for wall slip

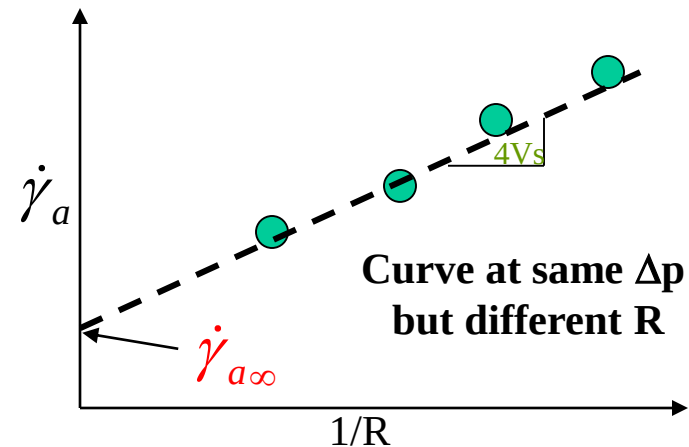
- Pick capillaries of different radii R
- Run experiment with each tube keeping Δp (i.e. τ_{wall}) same
- Plot app. shear rate vs $1/R$
- $\dot{\gamma}_{a\infty}$ is the corrected shear rate

$$\dot{\gamma}_a = \dot{\gamma}_{a\infty} + \frac{4V_s}{R}$$

Get viscosity from $\eta = \frac{\tau_{\text{wall}}}{\dot{\gamma}_{a\infty}}$

$$\dot{\gamma}_a = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{\text{wall}}} \right]$$

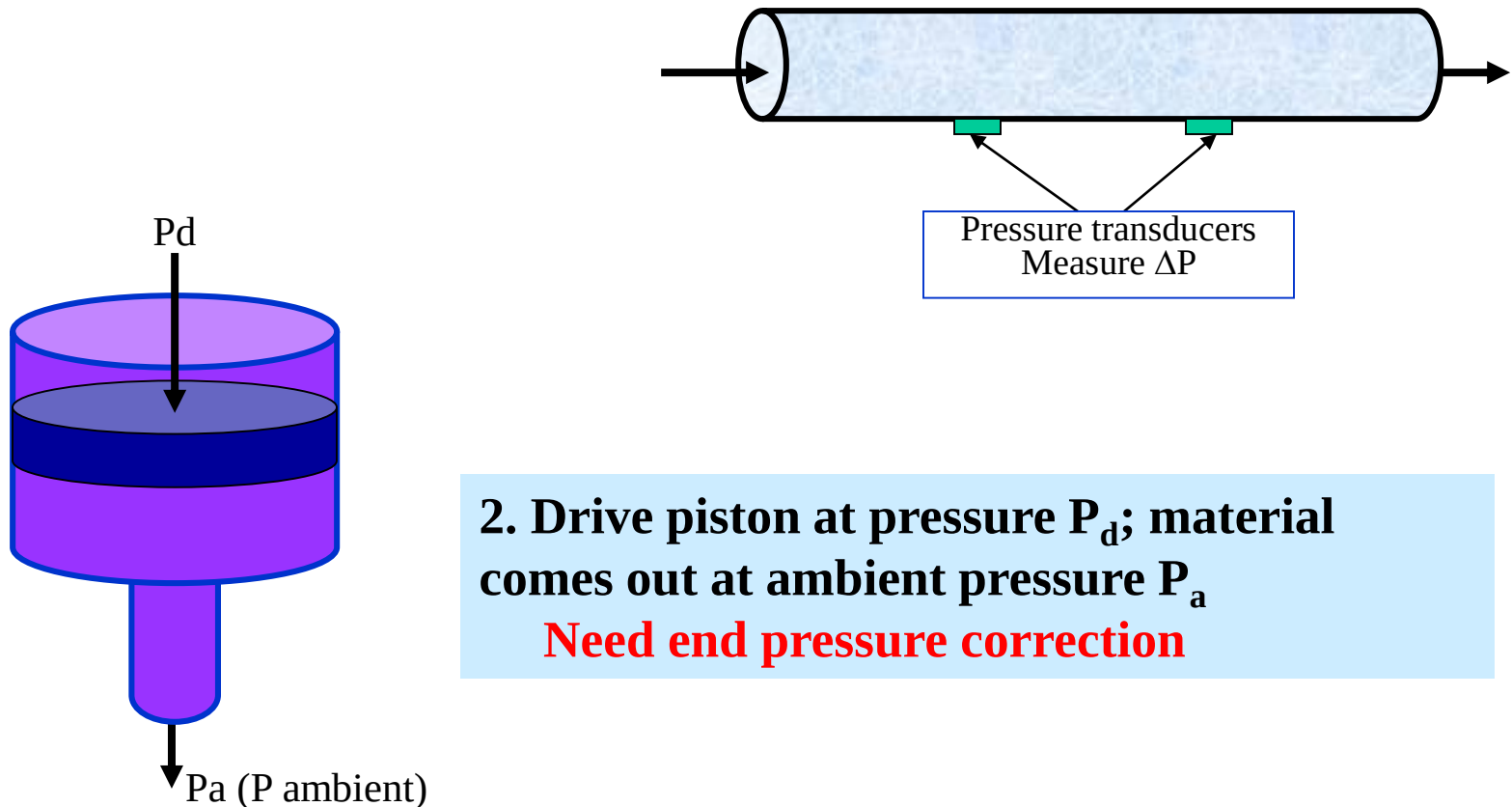
$\dot{\gamma}_a$ = apparent shear rate



- Repeat with different Δp to get different corrected shear rate

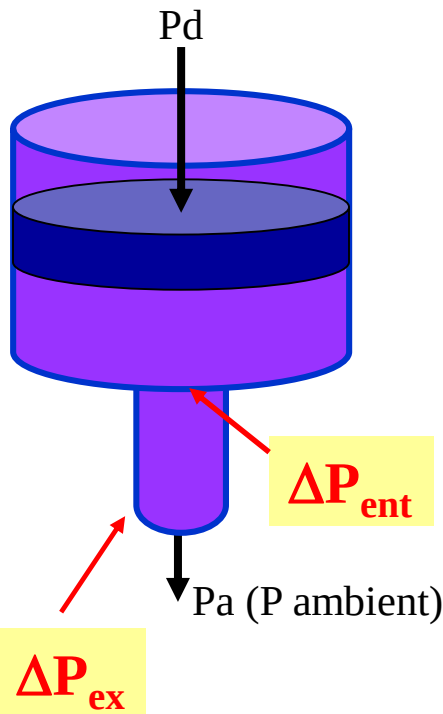
End Pressure Effects

1. Place pressure transducers far from the entrance and exit
No end pressure effect



2. Drive piston at pressure P_d ; material comes out at ambient pressure P_a
Need end pressure correction

End Pressure Effects



Measure $P_d - P_a$

But $P_d - P_a = \Delta P_{ent} + \Delta P + \Delta P_{ex}$

xs pressure from
entrance flow

pressure from flow

xs pressure from
exit flow

$\Delta P_{end} = \Delta P_{ent} + \Delta P_{ex}$ introduces errors and over predicts stress if not taken into account.
How?

Because measured $P_d - P_a$ often mistakenly taken as $\Delta P = 2\tau_w L/R$

End Pressure Effects-Bagley Correction

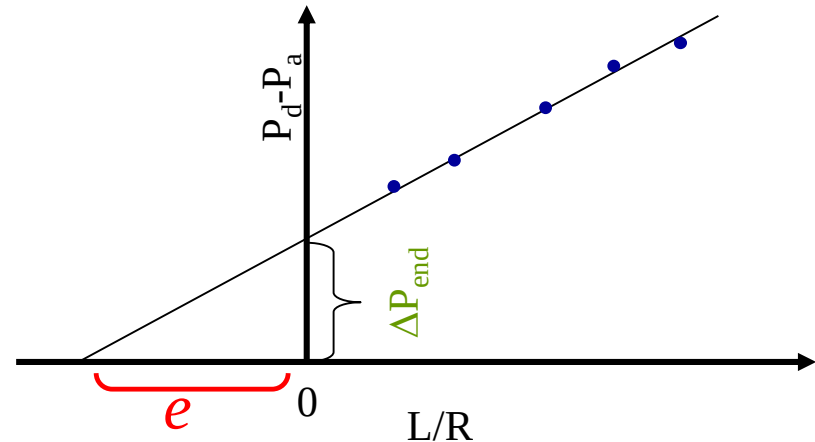
To correct for end pressure effect:

- Keep Q constant and measure P_d versus R (P_a known)
- Plot $P_d - P_a$ vs L/R
- From Q get shear rate
- Get τ from the equation

$$P_d - P_a = \frac{2\tau_w L}{R} + \Delta P_{end}$$

OR

$$\tau_w = \frac{P_d - P_a}{2(L/R + e)}$$



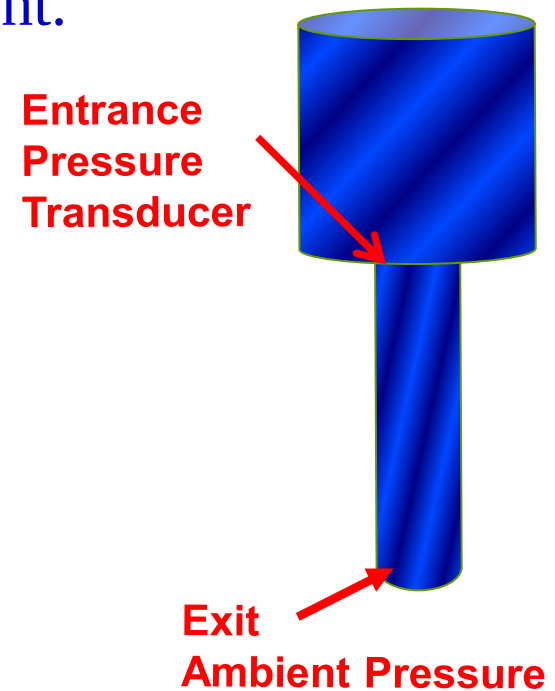
Data analysis

- AMS's lab in Venezuela 1999.
- Capillary pressure driven experiment.
- Polypropylene at 200°C.

•What do we correct for?

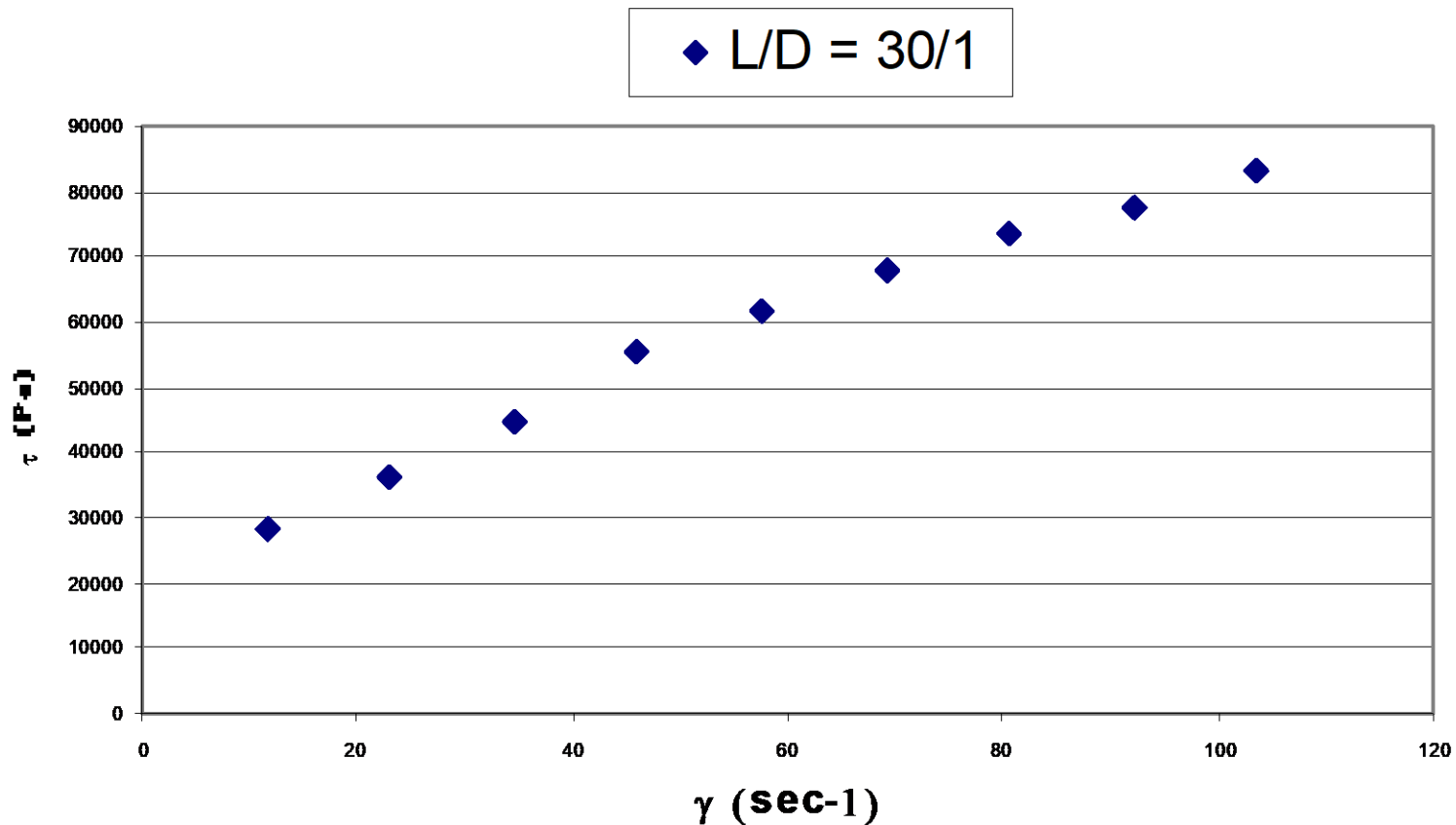
•End effects

•Non-homogeneous flow



Shear Flow Rheometry Experiment

Original Data: Position of Corrected Data?



Shear Flow Rheometry Experiment

- L/D ratios used:
 - 10mm/1mm
 - 20mm/1mm
 - 30mm/1 mm
- Corrections needed for stress τ_w and shear rate $\dot{\gamma}$.
 - Bagley plot:

$$\tau_w = \frac{P_d - P_a}{2(L/R + e)}$$

- Weissenberg-Rabinowitsch correction:

(For power law model)

$$\dot{\gamma}_{wall} = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{wall}} \right]$$

$$\dot{\gamma}_{wall} = \frac{Q}{\pi R_p^3} \left[3 + \frac{1}{n} \right]$$

Shear Flow Rheometry Experiment

- What are the Measured Variables?

–Flow Rate $Q = v_p \pi R_p^2$

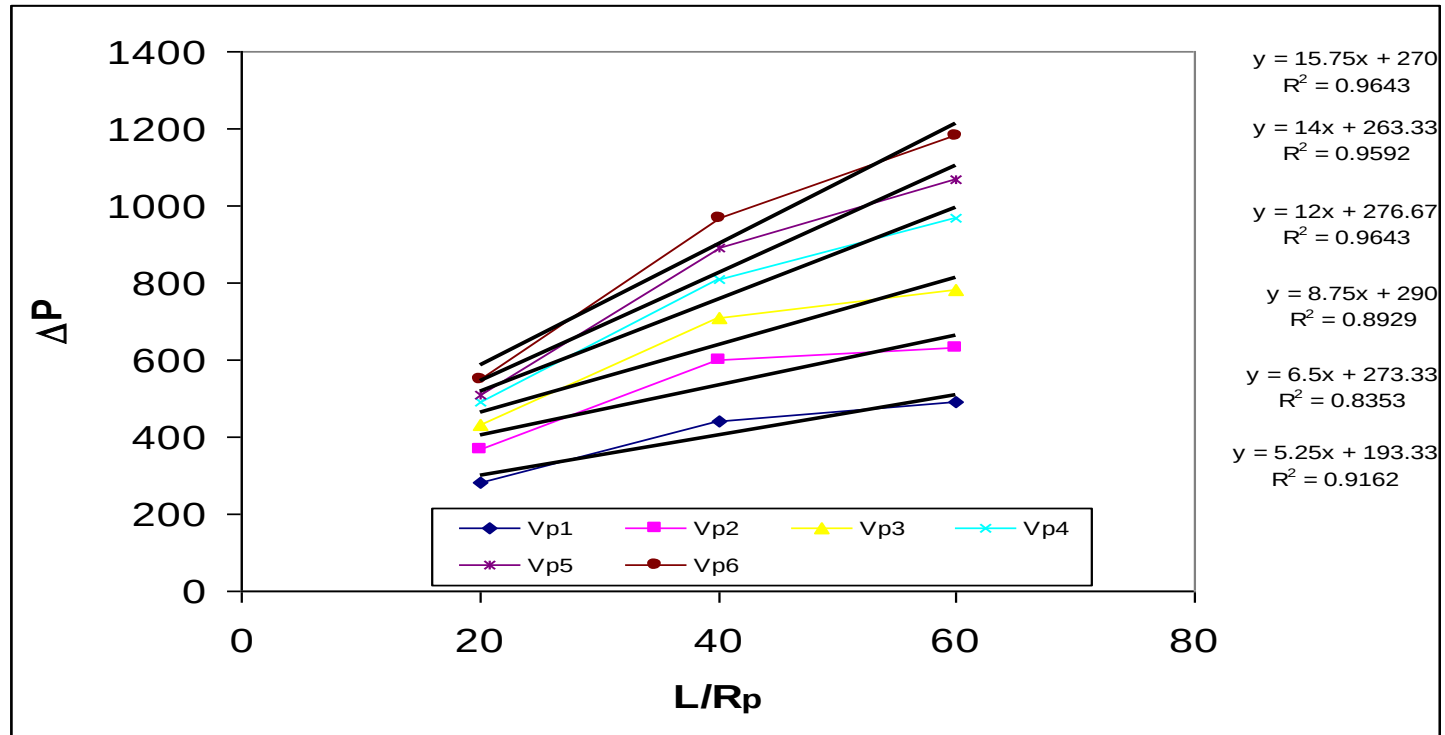
–Pressure Drop ΔP

Capillary: 30/1			Capillary: 20/1			Capillary: 10/1	
v_p (mm/s)	ΔP (psi)		v_p (mm/s)	ΔP (psi)		v_p (mm/s)	ΔP (psi)
0.01	490		0.01	440		0.01	280
0.02	630		0.02	600		0.02	370
0.03	780		0.03	710		0.03	430
0.04	970		0.04	810		0.04	490
0.05	1070		0.05	890		0.05	510
0.06	1180		0.06	970		0.06	550
0.07	1280		0.07	1020		0.07	580
0.08	1350		0.08	1070		0.08	620
0.09	1450		0.09	1110		0.09	640
1	4100		1	1170		1	670

Bagley Plot

- What do we plot?

➤ ΔP versus L/R_c



- Information obtained?

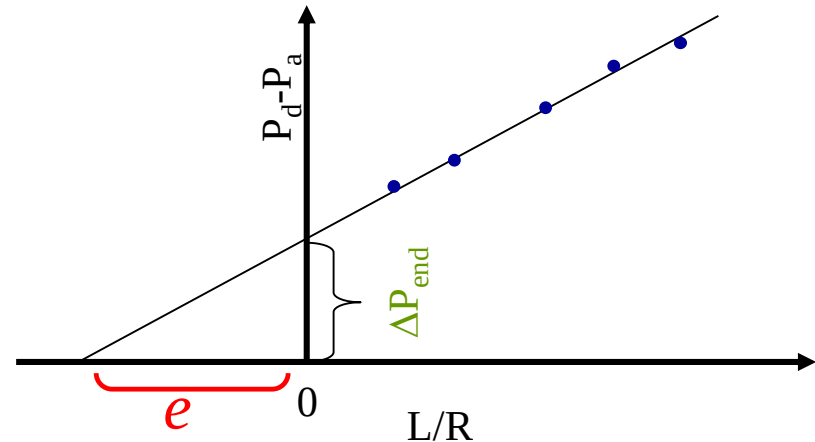
Bagley Plot

- Obtain Bagley Correction:

$$\tau_w = \frac{P_d - P_a}{2(L/R + e)}$$

- e = intercept/slope or when ΔP is zero:

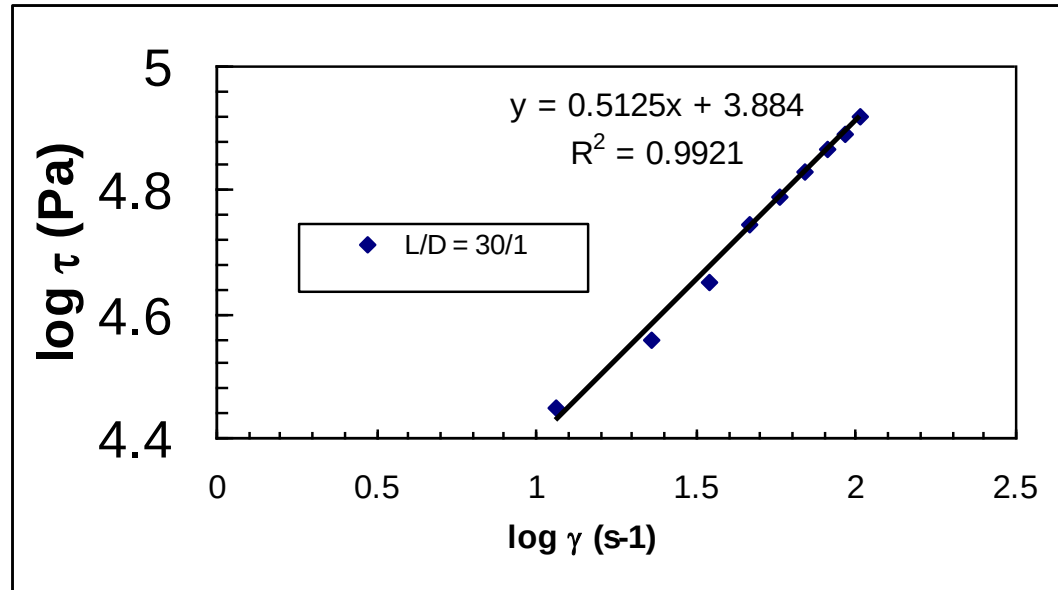
	x-intercept
Vp1	-17.1429
Vp2	-18.7857
Vp3	-23.0558
Vp4	33.1429
Vp5	-42.0508
Vp6	-36.7619
<Bagley factor>	



- Correct for stress τ_w with e as |<Bagley Factor>|.

Weissenberg-Rabinowitsch Correction

- Plot $\log \tau_w$ versus $\log \dot{\gamma}$.



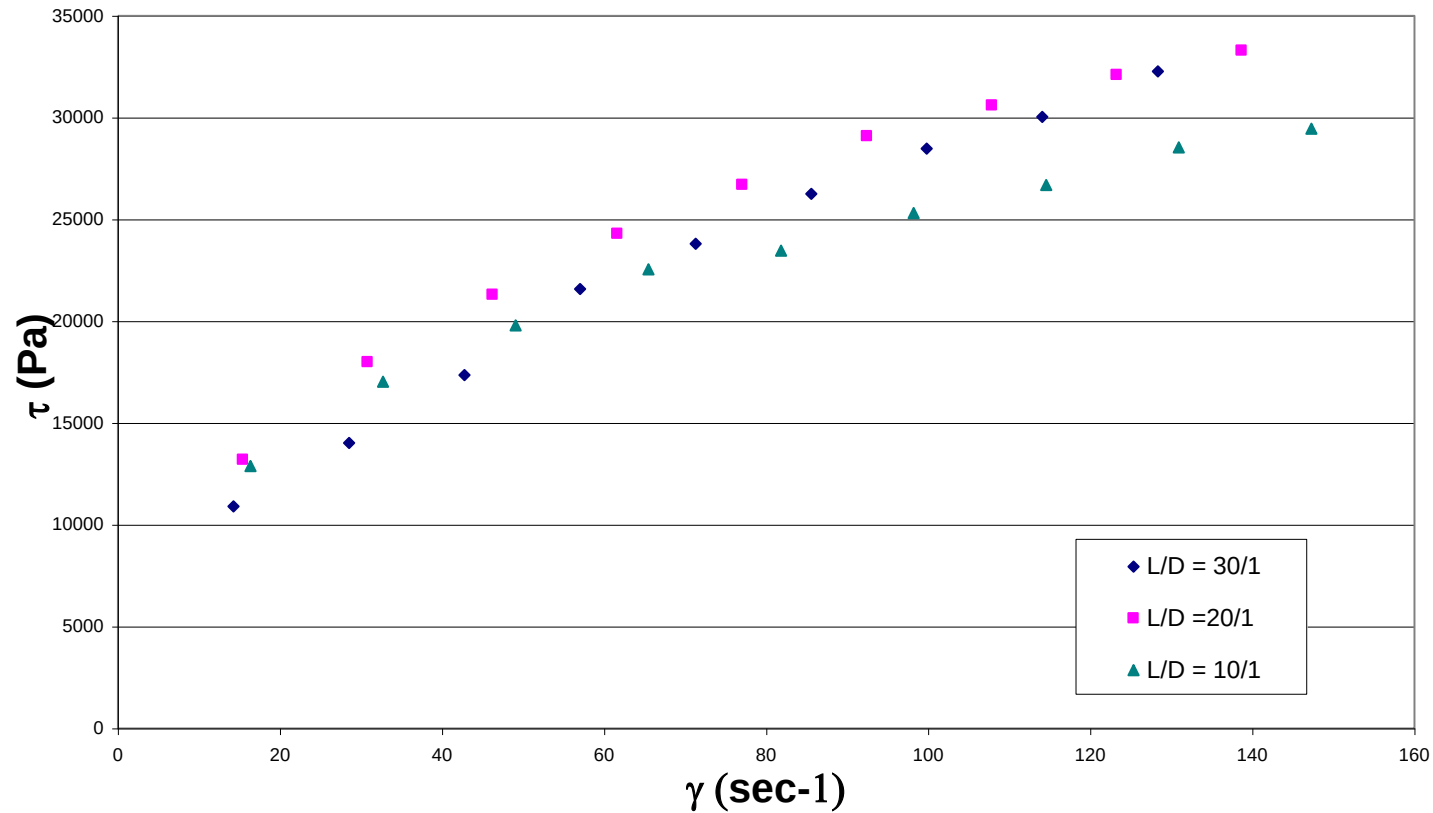
- Slope = n (for a power law model).
 - $n = 0.4256$ for 20/1 and 0.3728 for 10/1 L/D ratios

- Calculate the corrected shear rate $\dot{\gamma}_{wall}$

$$\dot{\gamma}_{wall} = \frac{Q}{\pi R_p^3} \left[3 + \frac{1}{n} \right]$$

Shear Flow Rheometry Experiment

Corrected Data



Shear Flow Rheometry Experiment

Comparison of Uncorrected data

